fun-miRBench Benchmark Report

Run setting	Value
Experiments evaluated	72
Experiments skipped	0
Predictors evaluated	5
Responsive genes	oriented shrunken log2FC > 0; repression FSOS s-value < 0.05 against d = 0.5
No-response genes	whole shrunken 95% interval of the oriented change below d = 0.5

Evaluation design

- Every predictor is scored on the same 19,226 genes; unscored genes rank last.

- Unresolved genes are left out of the AUROC.

- AUROC on 58 of 72 experiments (at least 30 responsive and 30 no-response genes).

- Spearman on all 72 experiments.

Key results

- Highest mean AUROC: miRBind2-3UTR (0.621).

- Highest mean Spearman: miRDB (0.075).

- Highest responsive pair coverage: miRBind2-3UTR (99.9%).

- Lowest responsive pair coverage: miRDB (25.0%).

Output tables

- per_comparison.tsv: every metric, for each experiment and tool.

- summary.tsv: means and 95% intervals, on all experiments and on unflagged ones.

- coverage.tsv: genes and pairs covered, and experiments retained, per tool.

- recovery.tsv: recall per experiment and tool, at each prediction budget.

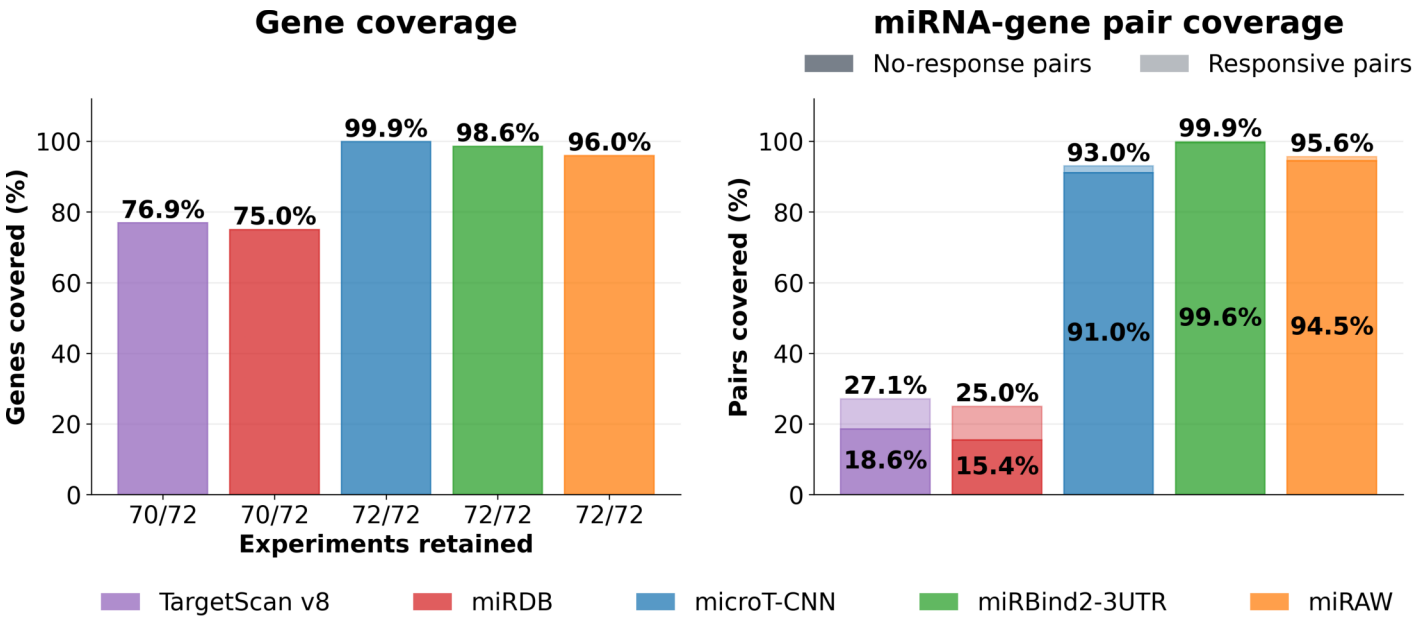
- precision_at_recall.tsv: precision and false positives at 50% recall.

Coverage

- Gene coverage: share of measured genes the tool scores in at least one experiment.

- Experiments retained: experiments where it scores responsive and no-response genes.

- Pair coverage: share of responsive and of no-response miRNA-gene pairs it scores.

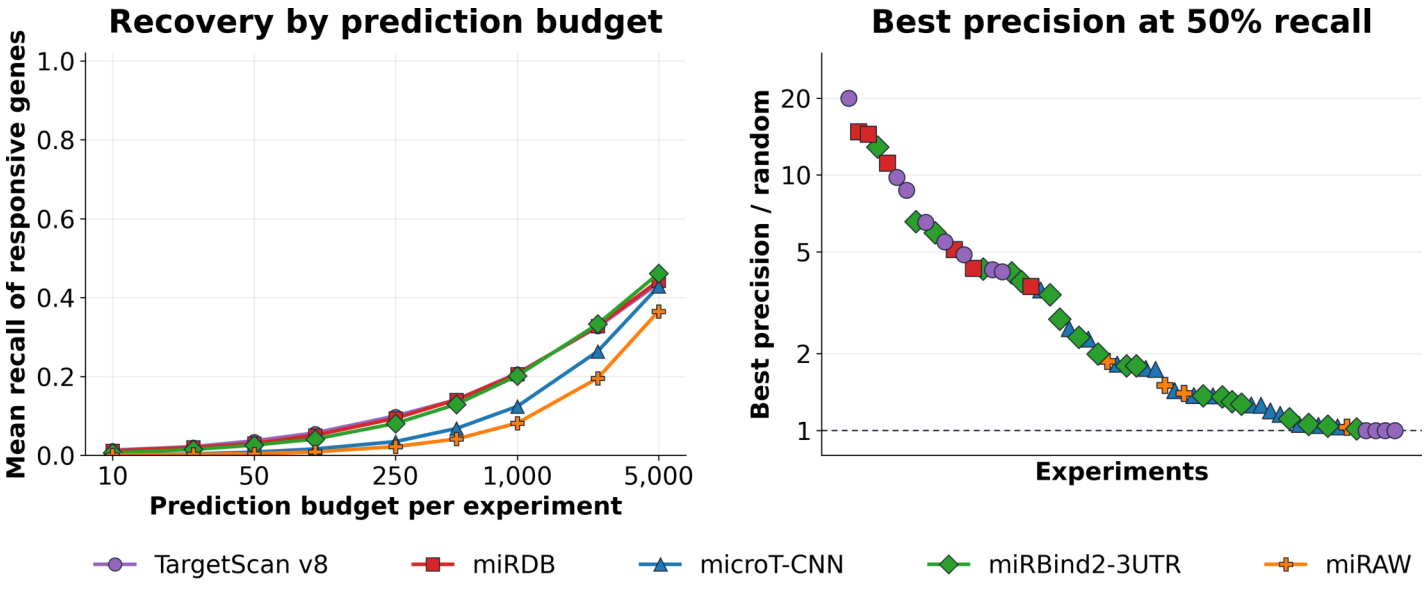


Recovery

- Recall: share of responsive genes in the top genes of the full ranking, mean over experiments.

- Best precision / random: best tool's precision at 50% recall over responsive share.

- Precision counts responsive and no-response genes only.



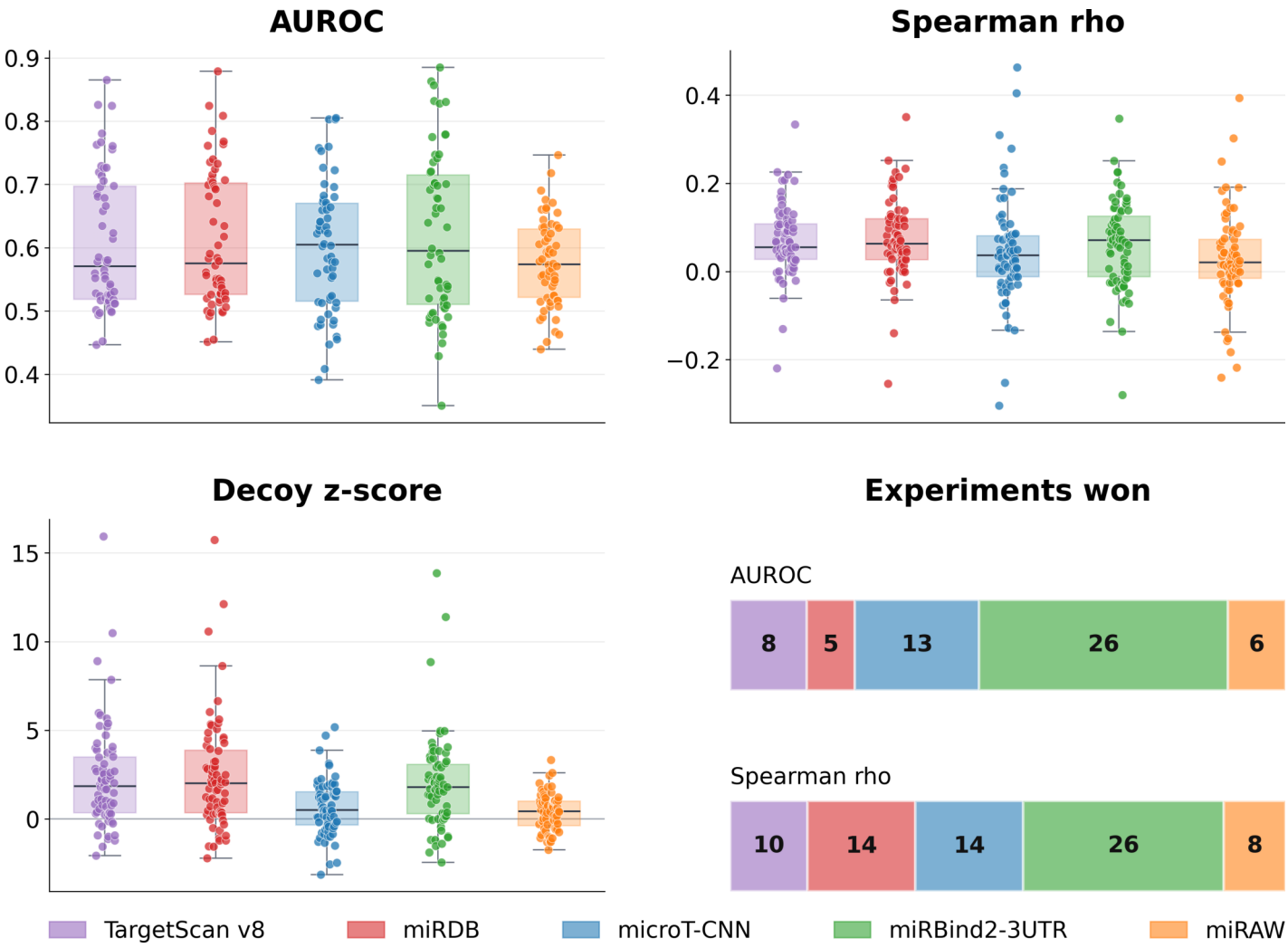
Performance

- AUROC: chance that a responsive gene scores above a no-response gene; 0.5 is chance.

- Spearman rho: agreement of the score's ranking with the shrunken log2FC; 0 is none.

- Decoy z-score: Spearman for the tested miRNA against different-seed miRNAs, in SDs.

- Experiments won: experiments where the tool has the highest AUROC or Spearman rho.



Coverage

Predictor	Gene coverage	Experiments retained	Responsive pairs	No-response pairs
TargetScan v8	76.9%	70/72	27.1%	18.6%
miRDB	75.0%	70/72	25.0%	15.4%
microT-CNN	99.9%	72/72	93.0%	91.0%
miRBind2-3UTR	98.6%	72/72	99.9%	99.6%
miRAW	96.0%	72/72	95.6%	94.5%

Performance

Predictor	Mean AUROC	Mean Spearman	Mean decoy z-score	Won, AUROC	Won, Spearman
TargetScan v8	0.608	0.069	2.280	8	10
miRDB	0.612	0.075	2.433	5	14
microT-CNN	0.601	0.046	0.645	13	14
miRBind2-3UTR	0.621	0.062	1.985	26	26
miRAW	0.577	0.030	0.371	6	8

Recovery

Predictor	Mean recall at 5,000 genes	Experiments won at 50% recall
TargetScan v8	0.436	12
miRDB	0.443	6
microT-CNN	0.428	16
miRBind2-3UTR	0.461	20
miRAW	0.365	4

Experiment inventory

Experiment	Experiment ID
hsa-miR-99b-5p · PC-3	E-MTAB-10350_hsa-miR-99b-5p_PC-3_24h
hsa-miR-138-5p · de-differentiated chondrocytes from osteoarthritic cartilage, osteogenic induction	GSE109108_miR-138-5p_DDC_day2
hsa-miR-138-5p · de-differentiated chondrocytes from osteoarthritic cartilage, osteogenic induction	GSE109108_miR-138-5p_DDC_day7
hsa-miR-574-5p · MGC-803	GSE112859_OE_miR_574_5p
hsa-miR-340-5p · A549	GSE115361_hsa-miR-340-5p_A549_24_h
hsa-miR-101-3p · HepG2	GSE115646_OE_miR-101-3p_HepG2
hsa-miR-133a-3p · HepG2	GSE115646_OE_miR-133a-3p_HepG2
hsa-miR-140-5p · HepG2	GSE115646_OE_miR-140-5p_HepG2
hsa-miR-18a-5p · HepG2	GSE115646_OE_miR-18a-5p_HepG2
hsa-miR-211-5p · HepG2	GSE115646_OE_miR-211-5p_HepG2
hsa-miR-1275 · DBTRG-05MG	GSE119161_hsa-miR-1275_DBTRG-05MG_record_48_h_series_summary_and_paper_day_3_72_h_paper_25_ug_per_35_mm_dish
hsa-miR-139-5p · CAL62	GSE123680_miR-139-5p_CAL62_96h
hsa-miR-150-5p · LX-2	GSE124411_OE_miR_150_5p_1
hsa-miR-375-3p · fibrolamellar carcinoma cell line (prFLC)	GSE125602_hsa-miR-375-3p_fibrolamellar_carcinoma_cell_line_prFLC_48_h_50_nM
hsa-miR-138-5p · 293T	GSE127502_miR-138-5p_293T_40nM_24h

Experiment inventory

Experiment	Experiment ID
hsa-miR-450a-5p · A2780	GSE129076_OE_miR_450a_5p_1
hsa-miR-126-3p · HMEC-1	GSE139947_miR-126-3p_HMEC-1_50nM_48h
hsa-miR-143-3p · HMEC-1	GSE139947_miR-143-3p_HMEC-1_50nM_48h
hsa-miR-126-3p · primary human lung microvascular endothelial cells, air control	GSE141129_hsa-miR-126-3p_HLMVEC_air_control_5_nM_16_h
hsa-miR-133a-3p · Flp-In T-REx 293	GSE142895_miR-133a-3p_T-REx-293_batch2_7d
hsa-miR-302a-3p · Flp-In T-REx 293	GSE142895_miR-302a-3p_T-REx-293_batch2_7d
hsa-miR-372-3p · Flp-In T-REx 293	GSE142895_miR-372-3p_T-REx-293_batch2_7d
hsa-miR-373-3p · Flp-In T-REx 293	GSE142895_miR-373-3p_T-REx-293_batch2_7d
hsa-miR-1271-5p · SH-SY5Y	GSE148827_OE_miR_1271_5p
hsa-miR-21-3p · HaCaT	GSE154968_OE_miR_21_3p
hsa-miR-330-3p · COV318	GSE158317_hsa-miR-330-3p_COV318_not_stated_paper_50_to_60_nM
hsa-miR-150-5p · primary human CD4+ memory T cells, activated 5 days	GSE161095_miR-150-5p_memoryT_input_24h
hsa-miR-20b-5p · HepG2	GSE168484_OE_miR_20b_5p
hsa-miR-100-3p · Kelly	GSE175729_OE_miR-100-3p_Kelly
hsa-miR-380-3p · Kelly	GSE175729_OE_miR-380-3p_Kelly
hsa-miR-485-3p · Kelly	GSE175729_OE_miR-485-3p_Kelly
hsa-miR-99b-5p · Kelly	GSE175729_OE_miR-99b-5p_Kelly

Experiment inventory

Experiment	Experiment ID
hsa-miR-101-3p · 143BTK- osteosarcoma cells in galactose medium	GSE180331_miR-101-3p_143BTK_120h
hsa-miR-130a-3p · bone-marrow mesenchymal stem cells	GSE202135_OE_miR_130a_3p
hsa-miR-130b-3p · bone-marrow mesenchymal stem cells	GSE202135_OE_miR_130b_3p
hsa-miR-139-5p · FUS H517Q/H517Q iPSC-derived motor neurons (paper and series); sample records say healthy	GSE203172_miR-139-5p_iPSC-MN_day34
hsa-miR-3648 · AGS	GSE204669_hsa-miR-3648_AGS_not_stated_qPCR_at_48_h_not_stated
hsa-miR-130b-3p · HUVEC	GSE204705_miR-130b-3p_HUVEC_48h
hsa-miR-375-3p · HUV-EC-C	GSE210778_OE_miR_375_3p
hsa-miR-1246 · Caco-2	GSE220728_ISO-miR-1246_G_Caco-2_5_isoform_100_pmol_mL
hsa-miR-1246 · Caco-2	GSE220728_ISO-miR-1246_a_Caco-2_5_isoform_100_pmol_mL
hsa-miR-1246 · Caco-2	GSE220728_hsa-miR-1246_Caco-2_canonical_100_pmol_mL
hsa-miR-17-3p · Ge518 patient-derived glioblastoma cells	GSE246708_hsa-miR-17-3p_Ge518_not_stated
hsa-miR-340-5p · Ge518 patient-derived glioblastoma cells	GSE246708_hsa-miR-340-5p_Ge518_not_stated
hsa-miR-138-5p · cartilage progenitor cells, chondrogenic pellets, 5% O2	GSE262978_hsa-miR-138-5p_cartilage_progenitor_cells_chondrogenic_pellets_5_O2_day_7
hsa-miR-1229-5p · Ishikawa	GSE267283_hsa-miR-1229-5p_Ishikawa_not_stated_not_stated

Experiment inventory

Experiment	Experiment ID
hsa-miR-637 · HPASMC in hypoxia (1% O2, 5% CO2)	GSE274053_hsa-miR-637_HPASMC_hypoxia
hsa-miR-661 · HPASMC in hypoxia (1% O2, 5% CO2)	GSE274053_hsa-miR-661_HPASMC_hypoxia
hsa-miR-369-5p · HMVEC-D, Lonza CC-2543	GSE293683_hsa-miR-369-5p_HMVEC-D_50_nM_24_h
hsa-miR-369-5p · HMVEC-D, Lonza CC-2543	GSE293683_hsa-miR-369-5p_HMVEC-D_50_nM_48_h
hsa-miR-526b-3p · MCF7 stable line, DMSO vehicle	GSE295502_hsa-miR-526b-3p_MCF7_stable_DMSO_48_h
hsa-miR-526b-3p · MCF7 stable line, doxorubicin 0.8 uM	GSE295502_hsa-miR-526b-3p_MCF7_stable_DOX_48_h
hsa-miR-526b-3p · MCF7 stable line, H2O2 216 uM	GSE295502_hsa-miR-526b-3p_MCF7_stable_H2O2_48_h
hsa-miR-655-3p · MCF7 stable line, DMSO vehicle	GSE295502_hsa-miR-655-3p_MCF7_stable_DMSO_48_h
hsa-miR-655-3p · MCF7 stable line, doxorubicin 0.8 uM	GSE295502_hsa-miR-655-3p_MCF7_stable_DOX_48_h
hsa-miR-655-3p · MCF7 stable line, H2O2 216 uM	GSE295502_hsa-miR-655-3p_MCF7_stable_H2O2_48_h
hsa-miR-451a · HL-60 stable, LPS 180 minutes in both arms	GSE296044_hsa-miR-451a_HL-60_LPS180min
hsa-miR-451a · HL-60 stable, untreated	GSE296044_hsa-miR-451a_HL-60_untreated
hsa-let-7a-5p · HEK293T	GSE303559_hsa-let-7a-5p_HEK293T_50nM
hsa-miR-126-3p · Human cord blood CD34+ CD133+ CD38+ progenitors	GSE78078_hsa-miR-126-3p_huCB-progenitor_CD133pos-CD38pos_72h

Experiment inventory

Experiment	Experiment ID
hsa-miR-126-3p · Human cord blood CD34+ CD133+ CD38- stem cells	GSE78078_hsa-miR-126-3p_huCB-stem_CD133pos-CD38neg_72h
hsa-miR-372-3p · U1810	GSE81417_hsa-miR-372-3p_U1810_30h
hsa-miR-1271-5p · MKN74	GSE87785_hsa-miR-1271-5p_MKN74
hsa-miR-1271-5p · SNU-601	GSE87785_hsa-miR-1271-5p_SNU601
hsa-miR-942-5p · OS-RC-2 renal cell carcinoma	PRJNA427217_hsa-miR-942-5p_OS-RC-2
hsa-let-7d-5p · HUVEC	PRJNA531758_hsa-let-7d-5p_HUVEC_not_stated_not_stated
hsa-miR-20a-5p · LX-2	PRJNA758728_hsa-miR-20a-5p_LX-2_not_stated_not_stated
hsa-miR-20a-5p · THP-1	PRJNA758728_hsa-miR-20a-5p_THP-1_not_stated_not_stated
hsa-miR-340-5p · LX-2	PRJNA758728_hsa-miR-340-5p_LX-2_not_stated_not_stated
hsa-miR-340-5p · THP-1	PRJNA758728_hsa-miR-340-5p_THP-1_not_stated_not_stated
hsa-miR-638 · SPC-A1 lung cancer line	PRJNA760808_hsa-miR-638_SPC-A1
hsa-miR-574-5p · HUVEC	PRJNA992921_hsa-miR-574-5p_HUVEC